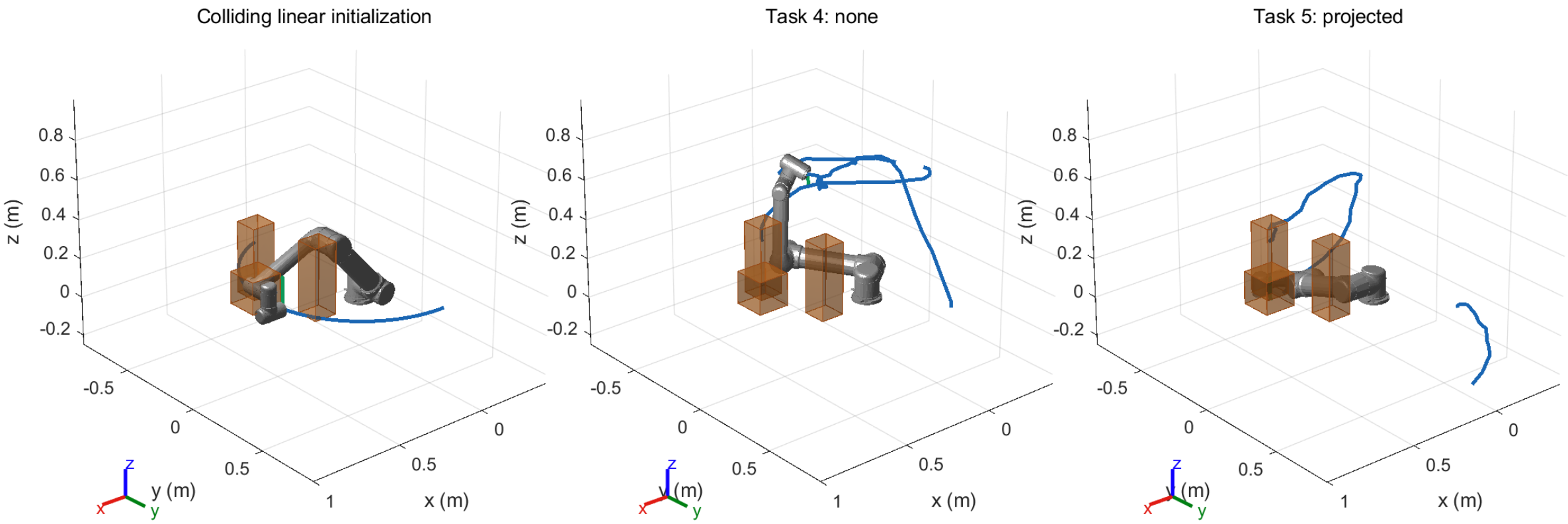
UR5 multi-obstacle transfer



Same upright endpoints and colliding linear initialization

Variant: side pillars move inward 40 mm and the right pillar becomes taller

Self-collision handling in V2

V1: all 20 projected paths were upright, but all self-collided

Tool y aligned with world +z, with yaw free.
V1 omitted self-collision from its objective.

V2: self-collision penalty and actual mesh rejection during updates

Feasible candidates take priority over lower cost.
Task 5 retains the PoE direction projection and orientation penalty.

Final gates: tilt <= 5 deg, proxy gap >= 35 mm, zero world and self collisions

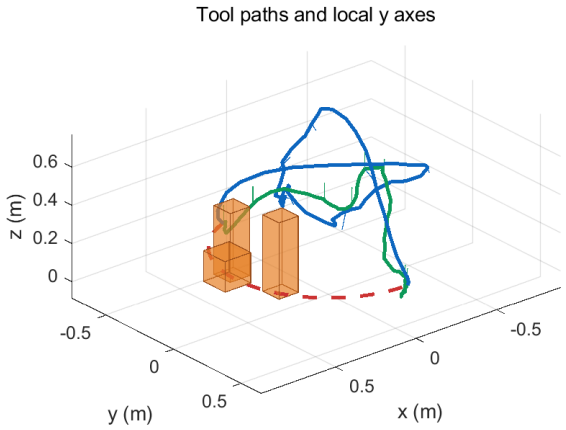
Fresh formal results

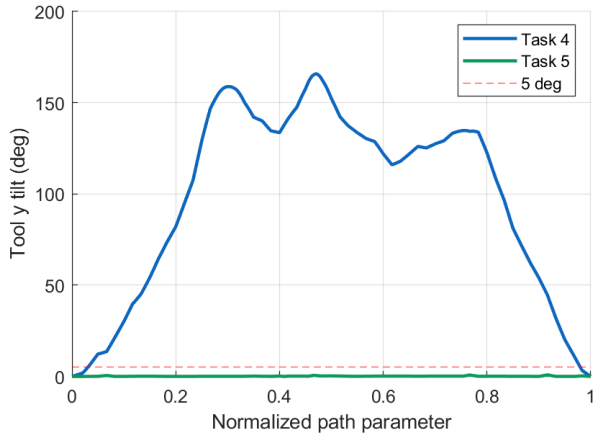
Scene / mode	Success	Median time (s)	Median max tilt (deg)
main / Task 4	10/10	30.53	126.55
main / Task 5	10/10	52.46	0.93
main / penalty only	0/10	301.48	-
variant / Task 4	10/10	29.83	134.52
variant / Task 5	10/10	48.44	0.95

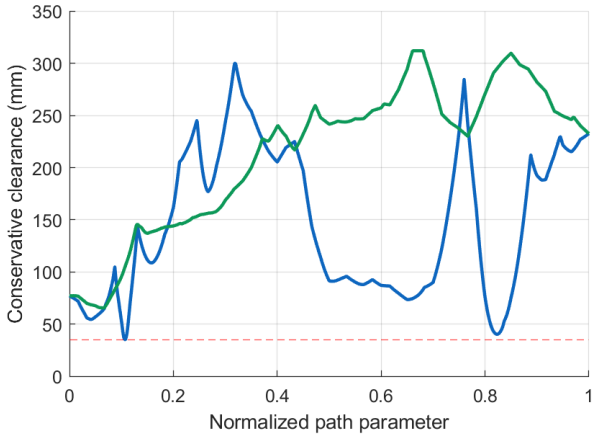
Common successful pairs: main n=10, variant n=10
Task 5 upright: 20/20. Self-collision failures: 0/20.

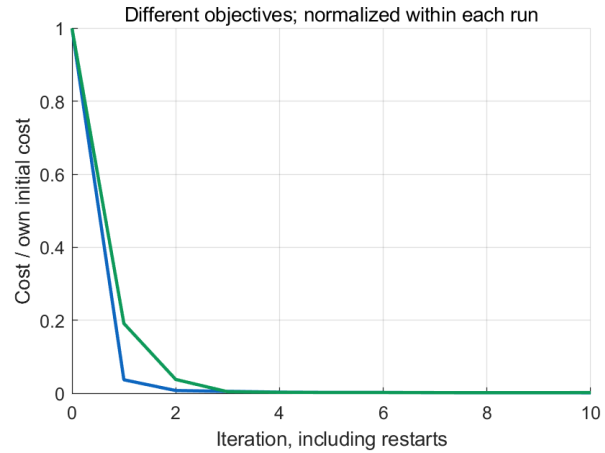
Fresh seeds 5200-5209. All failures counted. Time includes failures, tilt uses successes.

Member C contribution and limits









Added
Scenes, upright constraint, self-aware updates, fresh-seed study and replay

Reused
Task 1-3 PoE model, sphere envelopes and STOMP stages

Limits
Discrete checks, finite seeds, no payload or dynamics

Run: addpath scripts; run_task45_v2(...) Verify: verify_task45_v2(runRoot)